2024-25 First Semester
MATH1053 Linear Algebra I

Assignment 2

Due Date: Due on 24/Sep/2024 (Tuesday), at the end of Tuesday's lecture.

- Write down your **CHN name** and **student number**. Write neatly on **A4-sized** paper (*staple if necessary*) and **show your steps**.
- **Late submissions or answers without steps won't be graded.**

---

1. Write an augmented matrix for each of the following systems

(a). $\begin{cases} 2x_1 - x_2 = 3 \\ -4x_1 + 2x_2 = -6 \end{cases}$ (b). $\begin{cases} x_1 + 2x_2 + 3x_3 = 1 \\ x_1 + 3x_2 + 2x_3 = 1 \\ -x_2 + x_3 = 0 \end{cases}$

2. (a) Perform elementary row operations on the augmented matrices you obtained in Q1, in order to find an equivalent triangular system, respectively.
   (b) Based on the results of part (a), determine if the systems are consistent or not.
   (c) Solve the system based on your results in part (a).

3. Let A be an $n \times n$ matrix and let $\mathbf{x}$ and $\mathbf{y}$ be vectors in $\mathbb{R}^n$. Show that if $A\mathbf{x} = A\mathbf{y}$ and $\mathbf{x} \neq \mathbf{y}$, then the matrix A must be singular.

4. Let A be a square matrix.
   (a) Prove that $(A^2)^T = \left(A^T\right)^2$.
   (b) Use **mathematical induction** to show that $(A^n)^T = (A^T)^n$ for any positive integer n.

5. Let A and B be symmetric matrices. Show that $AB = BA$ **if and only if** AB is symmetric.

6. Let A be an $n \times n$ matrix. Show that if $A^{k+1} = O$, then $I - A$ is nonsingular and

$$(I - A)^{-1} = I + A + A^2 + \cdots + A^k$$

7. For the following system, use Gaussian elimination to obtain an equivalent system whose augmented matrix is in *row echelon form*. Solve for all possible solutions.

$$\begin{cases} x_1 & +2x_2 & -x_3 & = 1 \\ 2x_1 & -x_2 & +x_3 & = 3 \\ -x_1 & +2x_2 & +3x_3 & = 7 \end{cases}$$

8. For each of the following systems, use Gauss-Jordan elimination to obtain an equivalent system whose augmented matrix is in *reduced row echelon form*. Solve for all possible solutions.

(a). $\begin{cases} x_1 - 5x_2 = 6 \\ 3x_1 + 2x_2 = 1 \\ 5x_1 + 2x_2 = 1 \end{cases}$ (b). $\begin{cases} x_1 + 2x_2 + 3x_3 + 2x_4 + 15x_5 = 1 \\ 2x_1 + 4x_2 - x_3 + 2x_4 + 8x_5 = 6 \\ 3x_1 + 6x_2 - x_3 + 3x_4 + 13x_5 = 8 \end{cases}$

9. Let A, B, and C be

$$A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 1 & 2 \\ -1 & 2 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 2 & 0 \\ 0 & 1 & 2 \\ 1 & 2 & -3 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 4 & -3 \\ 0 & 1 & 2 \\ -1 & 2 & 0 \end{bmatrix}.$$

(a) Find an elementary matrix E such that $EA = B$.

(b) Find an elementary matrix E such that $EA = C$.

(c) Find an elementary matrix E such that $EB = A$.

(d) Find an elementary matrix E such that $EC = A$.

10. True or false. If false, give a counterexample. If true, explain or prove your answer briefly.

(a) If $A = A^{-1}$, then A must be equal to either I or $-I$.

(b) If A and B are singular $n \times n$ matrices, then $A + B$ is also singular.

(c) If A and B are non-singular matrices, then $(AB)^T$ is non-singular and

$$((AB)^T)^{-1} = (A^{-1})^T (B^{-1})^T$$

(d) A homogeneous system of linear equations must have at least one solution.

(e) A system of linear equations with fewer equations than variables always has at least one solution.

(f) Given A, B have same sizes, then $\text{rref}(A) + \text{rref}(B) = \text{rref}(A + B)$.